

backend/core/inference.py Documentation (Roman Urdu)

Yeh file backend aur local Ollama client ke darmiyan hone wali communication ya guftugu ka direct pull (bridge) hai.

File Ka Maqsad (Purpose)

Is mein direct HTTP (web) requests define ki gayi hain `httpx` ka plugin use karte huway jo background chalne wale port 11434 (jahan ollama zinda hai) ko orders bhejta hai aur json read/stream krta hai.

Code Blocks aur Functions ki Tafseel

Is file m bas ek hi main class hai: `OllamaClient`. OllamaClient app settings se automatically base URL leti hai.

1. Helper Functions

- `is_available()`: Ping marta ha check krny k liye connection theek ha.
- `list_models()`: Downloaded models (tags list) wapis karta ha json format me.
- `pull_model(name)`: UI se download / pull request any pe models ko gigabytes mein download karta hai aur bytes progress ko Frontend tak stream krta hai.
- `show_model(name)` / `delete_model(name)`: Model ki parameters/memory detail check karna ya usay remove karna.

2. `async def chat_stream(messages, model, temperature, ...)`

- **Kya karta hai?**: Yeh sabse ahem point ha. Jab user send push karta ha, chat streaming shuru hojati hai yahan se.
- **Logic**: Ye Max-Power mode setting bhi dekhta ha (agar enable hua to payload parameters m explicitly `num_gpu` pass kr dta ha Ollama ko force karnay k liye k saray model limits VRAM pr dalo). Phr stream shuru krk chunk-by-chunk event frontend bhejta hai `yield` generator bta k (`type: token` for alphaaz, `type: done` jab khatam hojaye).

3. `async def chat(...)`

- **Kya karta hai?:** Non-streaming normal function. Yeh single shot m baat behjta aur pura answer ikhatta return karta ha (agents context aur distillation verification me istmal houta ha yahan UI interaction nhen hoti isliye stream kiye baghir return kia gya hai).